
ISYE 6740 – Spring 2025

Project Proposal

Team Member Names:

Project Team 129

Ricky Wiryopranoto (GT ID: 903949033)

Ram Charan Singh (GT ID: 903860201)

Project Title: **Beyond the Pokédex: How AI Learns to Identify Species**

1. Problem Statement

Classifying animals into distinct taxonomic groups has long been a complex challenge in biology. Species identification relies on a combination of morphological traits, genetic data, and ecological factors, yet distinguishing closely related species or identifying organisms based on visual attributes alone remains difficult. Variability in colour patterns, shapes, and textures adds another layer of complexity, making manual classification prone to subjectivity and errors.

With advancements in data science and machine learning, automated classification methods have emerged as powerful tools for analysing visual characteristics and assigning categorical labels. Image classification models can learn patterns from large datasets and recognize subtle distinctions that may not be immediately apparent to the human eye. These techniques have been widely applied in fields such as biodiversity research, wildlife monitoring, and species discovery.

Inspired by these real-world challenges, this project explores whether machine learning models can accurately predict a Pokémon's typing based on its coloured image. Pokémon types, such as Fire, Water, and Grass, serve as predefined categories akin to biological classifications. By training different image classification models on Pokémon artwork, we aim to analyse how well visual features such as colour schemes, textures, and shapes, align with these categories.

Beyond Pokémon classification, this research has broader implications for biological taxonomy and species identification. If models can effectively map visual patterns to Pokémon types, similar methodologies could be applied to real-world animals. This could aid in species classification, discovering likenesses between different organisms, or even refining taxonomic groupings based on visual attributes.

Ultimately, this study seeks to evaluate the performance of different model architectures, assess their interpretability, and bridge the gap between fictional and real-world classification systems. The findings could offer insights into how visual features contribute to categorical distinctions in both artificial and real-world domains.

2. Proposed Methodology

To investigate whether machine learning models can predict a Pokémon's typing based on its coloured image, we will follow a structured approach involving data collection, exploratory data analysis (EDA) & preprocessing, model selection & training and evaluation.

Data Collection

We will be leveraging PokeAPI, an open-source API that provides extensive data on Pokémon including their types, attributes and images. The dataset will include Pokémon artwork and sprites along with their corresponding typings. For our project, we will be focusing on the first 7 generations of Pokémon.

EDA & Data Preprocessing

We will begin with Exploratory Data Analysis to gain a deeper understanding of the dataset. First, we will randomly select sample images to assess their quality, consistency, and suitability for model training. Next, we intend to visualize the distribution of different typings to verify for any class imbalances in the dataset. To further analyse visual characteristics, we will generate histograms to highlight common visual characteristics associated with each Pokémon type. Additionally, we will inspect for any missing or incomplete data entries in the Pokémon dataset and address them by either removing these entries or imputation based on other web sources. Lastly, we will vectorize each image and normalize the pixel values to ensure optimal preprocessing for the model.

Model Selection & Training

We will be testing both supervised and unsupervised machine learning models to evaluate their effectiveness in classifying Pokémon typings based on images.

Unsupervised Models

For unsupervised learning, we will be using clustering methods such as K-means and DBScan as well as density estimation techniques such as GMM to find natural groupings of Pokémon based on visual features extracted from the images.

Supervised Models

For supervised learning, we will be using techniques such as SVM, Random Forest and convolution neural network.

The dataset will be split into training and test sets following an 80-20 split. We will train models using categorical binary cross-entropy for multi-label classification.

3. Planned Evaluation Strategies

Each class of models will be evaluated separately.

For unsupervised learning models, we will be using PCA to reduce the dimensionality of the images to 2 dimensions, allowing us to visualize how Pokémon naturally cluster based on their visual features. This can help determine whether the groupings align with official typings or reveal unexpected insights. For instance, Charizard (Fire/Flying) might cluster closer to Dragon-type Pokémon like Dragonite rather than Flying-Types like Pidgeot, reflecting its dragon-like design despite its official classification.

For supervised learning models, each model will be evaluated using key classification metrics including accuracy & precision & recall evaluating the balance between correctly predicted typings and false typings, a confusion matrix to help visualize and understand how well different Pokémon types are distinguished and AUC-ROC curve to assess the ability of the model to distinguish between multiple types simultaneously.

We will also conduct feature importance analysis by plotting colour histograms and waterfall diagrams to evaluate the contribution of specific features to model performance.

Lastly, we will document insights gathered from wrong predictions to see if the model exhibits recurring misclassifications and identify potential reasons behind them across both supervised and unsupervised learning models. These insights will not only help to refine model accuracy but also provide a deeper understanding of how AI perceives and classifies visual images. This could provide guided improvements on preprocessing techniques, dataset balancing, or model architecture to better capture subtle but important distinctions.

4. References

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[3] Stiller, S., Dueñas, J. F., Hempel, S., Rillig, M. C., & Ryo, M. (2023). Deep learning image analysis for filamentous fungi taxonomic classification: Dealing with small data sets with class imbalance and hierarchical grouping.

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